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Promoting physical activity during recess in small sized elementary school: a pilot study

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Abstract

Background The aim of this pilot study was to examine the effect of structured and unstructured recess intervention based on spatial analysis technology on physical activity (PA) levels among children attending a small-sized elementary school.

Methods A small-sized elementary school (4.82 m^2 per child) was used as a pilot with 1162 students from grades 3 to 6. The effect of the intervention was assessed by a self-controlled experiment. Structured and unstructured recess intervention is the change of activity type and the addition of game markings based on spatial analysis. Duration and number of PA behaviors were measured with accelerometer and direction observation respectively.

Results During structured recess, post-tests demonstrated a significant increase in the duration of moderate to vigorous physical activity (MVPA) for both boys and girls ($p < 0.05$), as well as significant improvement in vigorous physical activity (VPA) for them ($p < 0.05$). Significant increases ($p < 0.05$) in the duration of VPA were shown among boys during unstructured recess; The number of students observed in the added markings spaces increased from 182 to 344; Significant increases ($p < 0.05$) in the proportion of MVPA were shown in the added markings spaces of the gate and playground.

Conclusions Activity type change to Tabata exercise improved PA during the structured recess of all students. Adding game markings based on spatial analysis technology increased participation number in PA during unstructured recess. This low-cost pilot intervention is expected to be used in the program of recess activities in small-sized elementary schools.

Keywords Physical activity, Recess intervention, School-based research, Space utilization, Spatial analysis, Environment strategies

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Background

It is well documented that physical activity (PA) can provide numerous physical and mental health benefits for children, with the most notable including motor development, cardiovascular and muscular health, maintenance of healthy weight and obesity levels, bone health, enhanced cognitive abilities, brain health, emotional regulation, and overall mood and quality of life [1, 2]. Despite these known benefits, the PA status of children and adolescents is still not optimistic [3]. For example, only 14.0% of children and adolescents in China meet the recommended amount of 60 min of moderate to vigorous PA per day [4]. The situation became increasingly severe after the COVID-19 pandemic, as numerous countries enforced government-mandated social distancing measures, resulting in a marked reduction in their overall physical activity levels [5]. Therefore, there is an urgent need to enhance the amount of PA among children in their daily lives.

Given that most children spend a significant portion of their waking hours in school, elementary schools serve as an ideal environment to encourage PA. In the school environment, recess can contribute up to 40% towards daily physical activity recommendations, and the overall contribution of recess and lunch breaks to daily physical activity can even exceed the daily recommended amount [6, 7]. Additionally, recess can be categorized as structured and unstructured. Unstructured recess refers to scheduled periods during the school day for free physical activity and play, while structured recess involves organized play, where games and physical activities are guided and led by a trained adult [8]. The World Health Organization (WHO) believes that recess time should be offered to all grade levels, and including both structured and unstructured recess [9]. It is apparent that PA interventions are necessary during both structured and unstructured recess. The school space serves as the physical medium that supports all PA. Previous studies have shown that the size of the available activity space for students is positively correlated with PA [10]. Therefore, regardless of the size of the school space, increasing the per capita available activity area from the perspective of spatial utilization may enhance children's PA levels during recess [11]. Especially in small sized schools. With the global urbanization process, the global urban population will grow by 2.2 billion people by 2050 [12], leading to limited construction land for existing urban central schools. This will pose challenges for land use amid the rapid changes in urban areas. As seen in rapidly urbanizing China, it will further reduce the per capita activity space for small-sized schools [13], whose area is below 10 m^2 per child [14].

School sports space utilization includes leveraging the school's open areas and the surrounding environment [15]. A system review studied the interventions that have been trialed to promote PA during recess in recent years [16]. Regarding structured recess intervention, the strategy primarily focuses on changing the activity type in the existing space. From the perspective of spatial utilization, on the one hand, this kind of intervention may guide students to use areas that were previously underutilized, such as parkour on the playground [17]. On the other hand, it may enhance the dynamism of activities by substituting space-intensive activities with those requiring less space, such as creating games [18]. However, not all studies reported on the intervention school size, suggesting that interventions from previous studies should be applied with caution in schools of different sizes, such as Parkour, which is difficult to conduct in small spaces. Regarding unstructured recess intervention, most of the studies chose the physical environment modification strategy in existing spaces [16]. On the one hand, this type of intervention can directly increase the per capita activity area by expanding the available activity space, such as adding sports facilities or providing activity equipment [16]. On the other hand, it can indirectly increase the activity space by defining the functional use of space, such as marking specific games [19]. Adding game markings belongs to the environment modification strategy, which has been recognized as a simple and low-cost way to intervene in child PA during unstructured recess [19]. Unfortunately, no studies have reported the specific site selection process for environmental modifications through spatial calculation, which is referred to as"unqualified physical environment modification". Given the PA is limited by time and space [20, 21], unquantified physical environment modification planning may face challenges such as limited access to facilities [22] and insufficient appeal of the available space [23], which in turn affects space utilization efficiency and ultimately undermines the effectiveness of the intervention. To ensure the intervention achieves its intended effects, it is essential to utilize spatial analysis techniques to support the design of recess activity plans.

A previous study created a topological network of the sports venues'road network, and calculated the area covered within a 15-min walk to the venues. According to the analysis result, the study pointed out that the location of future sports venues was distributed in the peripheral areas of the city, and the accessibility to venues in those areas would increase [24]. Another study used the integration value as the perceived accessibility indicator of playgrounds, and determined the attraction of different spaces through pedestrian flow simulation. The results found that playgrounds located on highly integrated streets tend to attract more people [25]. In recent years, there has been a growing integration of both types of accessibility analysis. A study integrated the calculation of integration with network analysis to analyze play spaces around refugee areas. The results showed that most informal play spaces are located on streets with good transportation access, which has potential for play. This was believed to be useful for decision-makers involved in the site selection of play facilities [26]. Currently, most sports facility site selection design is aimed at urban residents, with a focus on relatively large-scale urban spaces to design large sports facilities. However, only one study has intervened in unstructured recess from the perspective of spatial utilization, and it only addressed unstructured recess by reducing space population density (increasing the per capita activity area) [11].

The purpose of the study was: 1) to examine the effect of changing exercise type on students' PA in small-sized schools based on spatial analysis in structured recess. 2) to examine the effect of the game markings intervention during unstructured recess on students' PA in small-sized schools based on spatial analysis. The hypothesis of the study was: 1) Based on spatial analysis, changing the activity type will improve the PA levels of all students during structured recess. 2) Based on spatial analysis, adding game markings will enhance both the PA levels and the number of participants among all students during unstructured recess.

Methods

Study design

Based on the design of previous recess intervention research [16], a within-subject design was used to test the effectiveness of the intervention in this pilot study. Specifically, the study compared the differences between baseline data (May 24, 2023) and post-test data (after 1 week; May 31, 2023) from the case school. The time interval between the baseline and post-tests was consistent with previous studies [27, 28]. The accelerometer and SOPLAY observation tool were used in both baseline and post-test to evaluate physical activity duration and the number of active participants during recess ; the duration of the test was consistent with previous studies [29, 30]. On the days of the baseline and post-test, the average temperatures were 22 °C and 25 °C, respectively, and there was no recorded precipitation (0.00 inches) on either day. This study was approved by the Institution of Review Board at Shanghai University of Sport (Number: 102772022RT032), and additional approvals were obtained from the pilot school board and the president.

Pilot school selection

The study was conducted at an elementary school in Suzhou City, Jiangsu Province, China, which has a century-long history, and it served as the pilot school. The school has a total student population of 1,162 (students in grades 3–6), covering an area of only 5,611 m^2 (4.82 m^2 per child). The size of the pilot school is below most of the area standards [11, 14, 31], such as England (< 33m^2 per child), China (< 10m^2 per child), and America (< 80m^2 per child). According to the recess schedule, all students participate in both structured and unstructured activities recess simultaneously. The pilot school offers one structured recess per day in the morning for 15 min (Classroom: 5 min; Playground: 10 min), one unstructured recess per day in the lunch for 15 min, and all recess follows a 5-day school week (Monday–Friday). During structured recess, one homeroom teacher supervises students'participation in activities for each class every day. Additionally, two student representatives demonstrate on the podium, and all physical education teachers guide students in carrying out organized activities. During unstructured recess, the school does not provide any activity equipment. Except for areas with potential safety hazards (such as vehicle passageways), students have the freedom to participate in activities in spaces such as gardens, playgrounds, and other spaces, with no restrictions. Two designated supervising teachers will patrol the area for safety. The role of supervisors was solely to support the activity. If no safety risks were observed during the supervision, the supervisors did not provide any activity prompts to the students.

Participants

Prior to recruitment, a priori power analysis was conducted using G*Power 3.1 to determine the required sample size for detecting meaningful changes in PA behavior from baseline to post-intervention time points. Using an alpha of 0.05 and a power of 80%, the calculation of the power to detect a pre-post change in these measures was based on the effect sizes reported in a recent systematic review. It has been reported that the effect size is 0.79 [16], but due to high heterogeneity, a prudent estimation of the effect size (0.5) was used for the calculation. The 20% dropout rate was considered. Based on these parameters, the required sample was calculated to be 33.

All 3rd, 4th, 5th, and 6th-grade elementary school children had recess at the same time and were eligible to participate in the recess activities. All classes of the 3th,4th, 5th, and 6th grades were asked to participate in the study. In each class, all children were invited to participate in the recess intervention in the study. The exclusion criteria for participants include: (1) cognitive impairments, (2) ill or physically unfit to participate in PA, (3) injury during the study period, and (4) refusal to sign informed consent.

Intervention planing

Our intervention development was designed by referring to the common intervention strategies during recess [16]. Regarding the 15-min structured recess, due to the fixed area of the activity space each day, the content of the activities can be designed based on the size of the space. Tabata training is considered to have a small footprint, require less equipment, fit into the school environment, and improve athletic performance [32]. Moreover, such an intervention is free of cost and does not require any additional equipment. In the study, the intervention involved changing the type of activities, including incorporating Tabata exercises to replace the original activities. In the warm-up stage, martial arts exercises and broadcast gymnastics were replaced with 5-min Tabata warm-up exercises and 1-min intermittent small steps in place, the warm-up exercises included jogging, distance throwing, swimming, basketball, boxing, skipping rope, skating, and stretching imitation. In the exercise stage, 2 min of rope jumping were transformed into 4 min of Tabata exercise, with the specific exercises including backward and forward steps, high five between the legs, high leg lift, jumping jacks, and each movement cycle repeated 2 times. The content of the 5-min seated arm exercise in the classroom remains consistent with before the intervention (Table 1).

Regarding the 15-min unstructured recess, accessibility influences the frequency and duration of people's visits to a site, which is a key factor in the site selection design of physical environment modifications [25]. Accessibility analysis is divided into physical and perceived accessibility, aiming to emphasize physical distance and perceived distance, respectively [26, 33]. Thus, the intervention involves adding game markings, with the marking sites designed based on physical and perceived accessibility using spatial analysis (Table 1). The game markings contain various movements such as single-legged jumping, turning, curved running, and other actions that can develop balance, lower limb strength, and agility for students (Supplementary material Fig. 1).

Implementation

Our intervention implementation development was guided by the principle of school sports space utilization, which emphasizes the full utilization of the open space and the surrounding environment [15]. Prior to the intervention study, the researchers, the school principal, the director, the equipment room manager, and the physical education teacher participated in a 2-h discussion about the intervention design. The intervention plan and the corresponding intervention personnel were determined during the discussion. Arc GIS software was used to calculate the area of space, network topology analysis, and service area analysis (the maximum range that can be reached within a specified time via the existing road network). The Depthmap software was used to analyze the Integration value (IV) indicator and conduct Pedestrian Flow Simulation (PFS).

In the implementation of structured recess, the vector map of the pilot school was imported into ArcGIS 10.7 software to calculate the area of fixed exercise spaces (playground). The calculation results show the area was only 1.01 m^2 per child during structured recess, indicating that all students can only engage in activities in place. Therefore, after changing to Tabata exercise, the activities of structured recess were still carried out in the form of in-place. The intervention was led by a school PE teacher, and the researchers recorded teaching videos of structured activities for PE teachers.

Regarding unstructured recess, the study first considered distance accessibility to ensure that students could reach the game marking areas within the recess schedule time. Since the area for students'recess activities is self-chosen, the acceptable time to walk from the classroom to the farthest activity location was determined to be 1 min based on student self-reports (2 students from each grade). ArcGIS 10.7 software was used to construct the spatial plane network of the school and calculate the service area analysis for the 1-min reachable coverage,

Table 1 Intervention plan of structured and unstructured recess

Recess	Before intervention				Intervention			
	Content	Frequency	intermittent	Duration	Content	Frequency	intermittent	Duration
Morning Recess*	**A**:Seated arm exercise in Classroom	1t	0	5 min	**A**:Seated arm exercise in Classroom	1t	0	5 min
	A:Radio drill in Playground	1t	0	5 min	**A**:Tabata warming up in Playground	20 s/t	0	5 min
	A:Military drill in Playground	1t	0	3 min	**A**:Intermittent small steps in place	1t	0	1 min
	T:Boys and girls take turns jumping rope in the playground	1 min/t	1 min	2 min	**A**:Tabata training in Playground	20 s/t	10 s	4 min
Lunch recess#	**A**:Free activity without any sports equipment in school spaces	1t	0	15 min	**A**:Free activity with game markings in school spaces	1t	0	15 min

t, time; *, structured recess; #, unstructured recess; T, Boys and girls take turns participating; A, all students participating forming an intersection area as the maximum design range for the game marking locations, as shown in Fig. 1 and Supplementary material Fig. 6 .

The study next focused on how perceived accessibility indicators might influence the intervention effect in site selection design, specifically based on the relationship between integration value (one of the perceived accessibility indicators) and its potential to attract more pedestrian flow [25]. The study planned to add markings to the negative space (areas with relatively low space usage frequency) and the active space (areas with relatively high space usage frequency), within the maximum design range. The study analyzed the integration value (IV) to identify alternative negative and active spaces. Considering that pedestrian flow simulation (PFS) reflects the frequency of space usage [25], the spatial flow was subsequently verified through PFS, and the markings were finally placed in the selected areas. All of the above analyses were conducted using Depthmap X software. In detail, according to the integration value, a lower value is more likely to indicate a negative space, and the bottom 50% of areas, ranked from high to low, are considered negative spaces (Fig. 1). During the flow simulation, assuming an average class size of 40, 40 agent robots are released in each classroom. The trajectory of activities within 15 min was observed to validate the correctness of the selection of target-added markings space. Spaces with higher pedestrian flow (the redder the color) are identified as active spaces. The study identified space Z1 (IV ≈ 0.35; top 50%) and space Z2 (IV ≈ 0.33; top 50%) as activity spaces within the 1-min maximum design range, while space Z3 (IV ≈ 0.29; bottom 50%) and space Z4 (IV ≈ 0.28; bottom 50%) were classified as negative space accordingly (Supplementary material Fig. 6), and these spaces were selected as targets added markings space (Fig. 1). During the process of adding, existing school equipment was utilized as marking materials, such as existing rope, agile circle, and other moveable equipment. The teaching director informed students about the change in the activity space through the whole school broadcast before recess.Besides the intervention recesses (structured and unstructured) mentioned above, all other intervention settings remain unchanged.

Data collection

Ancillary measures

The date of birth and sex of the participants were obtained from the school principals. A cohort of students in grades 3–6 completed anthropometric measures as part of an ancillary study. The body height and weight of children were measured using height and weight measuring instruments (HKJY, HK6000-ST), and BMI was calculated from these values.

Duration of different physical activity

The study used a stratified random sampling method to select 40 accelerometer wearers based on before selecting the method [34]. Specifically, 10 students (5 boys and 5 girls) were randomly selected from each grade, using student IDs as the sampling frame. The selection process was carried out using the random number generator feature in SPSS (Version 23.0). The last two digits of the randomly generated numbers were matched with the corresponding student IDs to identify the participants. The Actigraph GT3X + accelerometer has previously been validated for measuring PA in elementary school children [35]. Participants ($n = 40$, grades 3 to 6) wore the Actigraph GT3X + accelerometer from 9:00 AM to 15:00

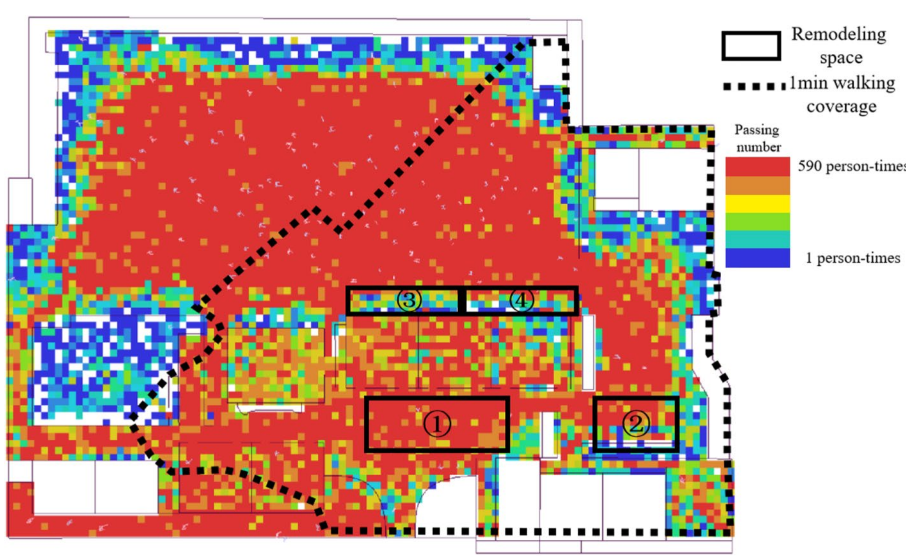

Fig. 1 The max marking range and agent analysis result

PM during both baseline and post-test days. The accelerometer was worn on the right hip and secured by an elastic waist belt. The researchers demonstrated how to wear an accelerometer in the classroom. The duration of light physical activity (LPA), moderate-to-vigorous physical activity (MVPA), and vigorous physical activity (VPA) were assessed by accelerometer (30 Hz, 10 s epoch). The study scores the accelerometer data using ActiLife version 6.0 software. Considering that the measurement target is elementary school children in China, the intensity of physical activity (PA) was classified based on the physical activity intensity criteria for Chinese children into three categories: LPA (101–2800 counts/min), MPA (2801–4000 counts/min), and VPA (4001 counts/min) [36]. Children included in the analysis must have at least one baseline recess and one intervention recess data, with each recess period consisting of both 15-min structured recess and unstructured recess.

Number of physical activity behavior

Regional level PA intensity, PA type, and play environment were measured using the Youth Play and Leisure Activity Observation System (SOPLAY) [38]. Video-assisted direct observation can improve the reliability of direct observation and measurement [38]. In the study, since structured recess did not alter the activity locations, while unstructured recess allowed for more freedom in school spaces, it was necessary to observe how students utilized the available space during unstructured recess. Unstructured recess videos of each identified target area were recorded through each data collection process. The observed subjects included all the students participating in the intervention at the school. SOPLAY instruments are utilized to analyze each recess video, counting the number of students engaged in sedentary behavior, walking, or vigorous activity behavior. Active activities classified as MVPA excluded static behaviors [38]. The activity categories were recorded by gender, with observers determining gender based on the children's dress and hairstyle to the best of their ability.

The physical activity data with an observation interval of 2 min showed the highest consistency with directly measured PA levels [38]. In the study, observation was scanned at 2-min intervals (a total of 7 times), with two trained assessors simultaneously scanning the same area. To clearly observe the large size of several play spaces, the added markings garden (Z1) were respectively subdivided into two areas. The added markings space at the school gate (Z2) and playground (Z3; Z4) were treated as an area respectively (Supplementary material Fig. 7). All subdivisions obeyed the SOPLAY protocol [39].

The observers underwent a two-day training. On the first day, they reviewed the SOPLAY protocol for 2 h, familiarizing themselves with activity classification standards, scanning techniques (left to right, girls followed by boys), and scanning speed (1 child per second). On the second day, the observers received training on properly calculating PA counts using the developer's online instructional videos for 2 h [40]. Afterward, the reliability of behavior observations was assessed through joint observation by the two observers for 2 h. To assess the reliability of the observation, video recordings of the activities of 9–12 years old children during recess were utilized. The reliability test involved assigning two assessors to the same set of 100 observations, with each assessor scanning four same areas (25 scans per area). In cases of large discrepancies between assessors ($r < 0.70$), the time point was reanalyzed. Intra-class correlation coefficients were calculated at r = 0.88 (95% CI: $r = 0.85 \sim 0.90$) between assessors.

Statistic analysis

In the statistical process, average and standard deviation were used to analyze the duration of PA of different intensities during recess. All kinds of PA behaviors in each activity space were summed up separately. The total number of students in each activity space was also recorded. For the intervention effect analysis, a two-tailed paired sample T-test respectively compared the differences between baseline and intervention in the duration of static LPA, MVPA, and VPA by sex and overall. Additionally, chi-square tests of independence were performed to assess the statistical significance of changes in the proportion of school students participating in each type of PA behavior during baseline and intervention. All statistics were analyzed using SPSS Version 23.0, and the statistical significance was set as $p < 0.05$.

Results

Physical characteristics and demographic measures

Participants were selected from the entire school to receive the intervention, including 589 boys and 552 girls. A total of 40 preschool-aged children were selected as accelerometer subjects. The sample size is sufficient to detect an intervention effect and exceeds the sample sizes used in previous recess intervention studies [27]. Descriptive information for a cohort of participants was reported in detail (Table 2).

Outcomes of structured recess

Descriptive measures for PA in boys and girls separately from baseline to the intervention were reported in Fig. 2. Overall, the duration of LPA for students significantly decreased from 9.05 min to 6.45 min ($p < 0.05$, 95%CI: $-3.60 \sim -1.60$) during 15 min structured recess, while the duration of MVPA increased from 3.20 min to 5.51 min ($p < 0.05$, 95%CI: $0.94 \sim 3.66$), and the duration of VPA increased from 1.55 min to 3.66 min ($p < 0.05$,

Table 2 Physical and demographic characteristics of participant

Items	All participants (Mean ± SD)		Accelerometer Subject(Mean ± SD)	
	Boys($n = 589$)	Girls($n = 552$)	Boys($n = 20$)	Girls($n = 20$)
Age(year)	10.2 ± 1.2	10.1 ± 1.2	10.4 ± 1.3	10.3 ± 1.3
Height(cm)	148.01 ± 10.0	147.9 ± 10.5	149.4 ± 10.7	151.5 ± 11.4
Weight(kg)	44.5 ± 12.9	41.0 ± 10.9	40.7 ± 7.5	41.8 ± 9.6
BMI(kg/m^2)	20.0 ± 3.9	18.4 ± 3.0	18.10 ± 1.8	18.04 ± 2.5

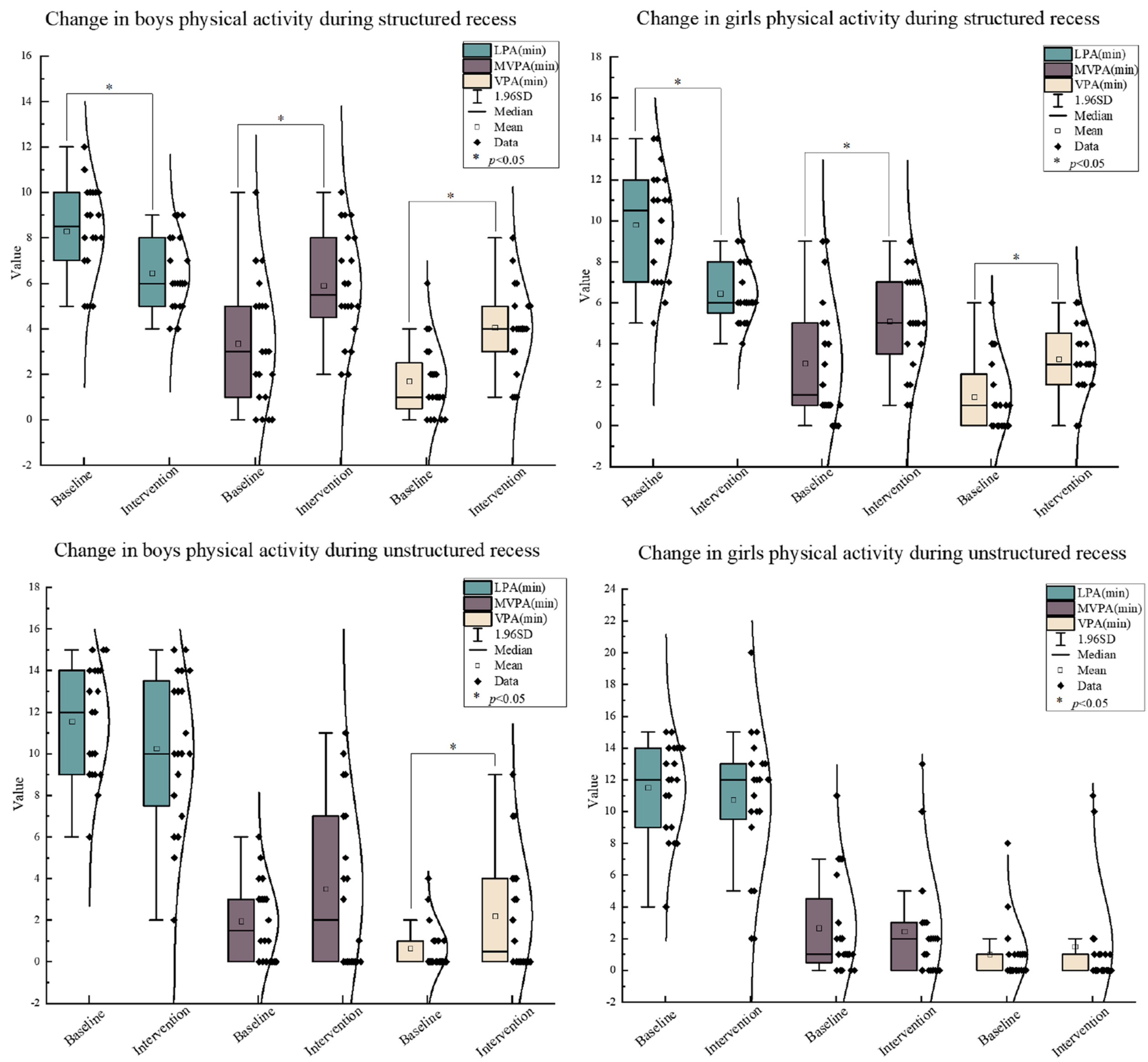

Fig. 2 The changes of duration of physical activity from baseline to intervention

95%CI: 1.23 ~ 2.97). The baseline differences in PA level during recess were not statistically significant by gender. For both boys and girls, post-intervention demonstrated a significant increase in the duration of MVPA (boys: $p < 0.05$, 95%CI: 0.51 ~ 4.59; girls, $p < 0.05$, 95%CI: 0.067 ~ 4.03), as well as significant improvement in VPA (boys, $p < 0.05$, 95%CI: 1.08 ~ 3.62; girls, $p < 0.05$, 95%CI: 0.56 ~ 3.14). The results for PA in both boys and girls revealed a significant decrease in the duration of LPA (boys, $p < 0.05$, 95%CI: −3.32 ~ −0.38; girls, $p < 0.05$, 95%CI: −4.74 ~ −1.96).

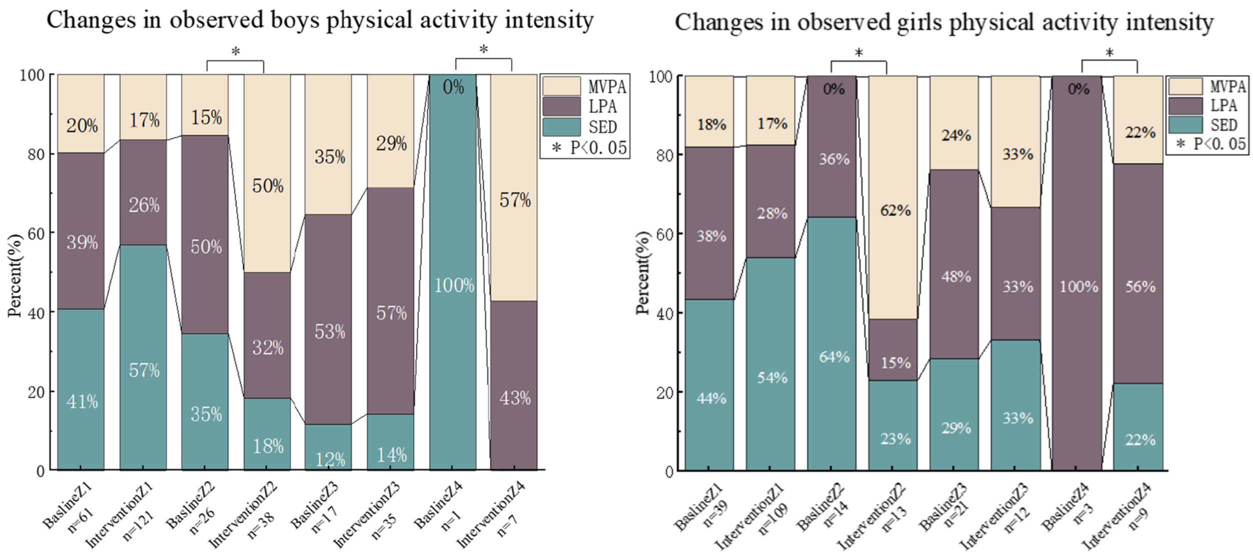

Fig. 3 The changes of physical activity behavior from baseline to intervention

Outcomes of unstructured recess

The baseline differences in PA level during 15-min unstructured recess were not statistically significant by gender (Fig. 2). Overall, the duration of LPA for students decreased from 11.53 min to 10.50 min ($p > 0.05$, 95%CI: −2.50 ~ 0.45), while the duration of MVPA increased from 2.30 min to 2.98 min ($p > 0.05$, 95%CI: −0.78 ~ 2.13), and the duration of VPA increased from 0.83 min to 1.85 min ($p > 0.05$, 95%CI: −0.082 ~ 2.13). Significant increases ($p < 0.05$, 95%CI: 0.11 ~ 2.98) in the duration of VPA were shown among boys after the intervention.

The number and proportion of PA behavior from baseline to intervention are presented in Fig. 3. Intervention demonstrated a positive effect, with the total number of students observed in the added markings spaces increasing from 182 to 344 during unstructured recess. After the intervention, the total number of MVPA students increased from 35 (19.23%) to 87 (25.29%), with this increase being statistically significant ($p < 0.05$). Specifically, the number of MVPA in Z2 and Z4 increased by 23 and 5, respectively. The findings indicate a significant increase in the proportion of MVPA in Z2 ($p < 0.05$) and Z4 ($p < 0.05$). However, there were no significant increases in the proportion of MVPA in Z1 ($p > 0.05$) and Z3 ($p > 0.05$).

Discussion

Based on the existing recess intervention strategies, the study changed the exercise type to Tabata exercise for the structured recess and added game markings for unstructured recess based on spatial analysis. The purpose of the study was to examine the impact of interventions on recess PA in small-sized elementary schools. The main finding supported the first hypothesis, indicating that changing the exercise type to Tabata can enhance PA during structured recess in school. The intervention results partially supported the second hypothesis, demonstrating that added game markings based on spatial analysis for unstructured recess increased the number of participants engaging in PA.

Structured recess

In the study, the structured recess intervention focused on improving students'PA through in-place activities, aiming to achieve the intervention goal by modifying the activity content. Although some studies have introduced Tabata training into elementary school intervention, and confirmed its effectiveness in promoting high-intensity exercise within a school environment [41]. Research comparing the effects of Tabata and other exercise models on physical activity outcomes when introduced into primary school activity programs remains limited. Until now, most studies have primarily focused on university student populations. The findings were consistent with previous studies, indicating that Tabata training can increase physical activity in students'activity programs [42]. One possible explanation is that the study changed the type of activity to reduce the space occupied by the equipment during the activity, which involved changing rope jumping to freehand actions. This approach is similar to previous studies, expanding the average activity area required may increase the amount of PA [11]. Another possible explanation is that the study reasonably designed the activity content. It is crucial to note that a single in-place action may reduce the activity intensity of exercisers due to low interest [43]. Thus, the study designs Tabata movements that involve multiple classical exercise movements, which have been proven to arrive at high-intensity levels [44, 45]. Moreover, most studies ignored the design of the warm-up before the Tabata exercise [46]. The study added warm-up movements with Tabata exercise principles [47], which could fully prepare for the following high-intensity exercise.

Unstructured recess

In terms of unstructured recess, the study specifically focused on the intervention effects of game markings in different spatial layouts and designed the intervention using spatial analysis techniques, providing a new perspective for utilizing school space during recess. Up to now, no studies have been reported on recess intervention studies similar to those size area schools. Despite variations in school size, the study is similar to the positive results of the previous intervention to add game markings in large-size schools [48]. The results of the study prove the effectiveness of applying the principles of school sports spatial utilization [15] and the usefulness of applying in a small school with game markings [11]. The potential explanation lies in the use of spatial analysis technology, which calculates the accessibility spaces and pedestrian flow between classes within a fixed time. Through the service area analysis, game markings create more reachable sports functions in negative and active spaces. This approach increases the chance of students utilizing school space, all while ensuring their accessibility within a limited time frame.

Different from the original hypothesis, the proportion of MVPA and the number of people in the gate and playground space increased significantly, while the situation in the garden areas (active space) did not change significantly from the baseline. A study reported that certain differences in the intensity of PA were observed in different regions with different environmental modifications [41]. Although the result of the study is similar to the prior research, the same type of game marking strategy is used to carry out intervention in different spatial layout areas. The possible explanation of the study is the population density, an excessive influx of people into a small space can also limit physical activity [10]. In the study, the distance between the classroom and garden areas is relatively close, and the integration value and pedestrian flow simulation indicated the potential to attract a large volume of people. Most of the students choose to go to the garden for recess, proximity to garden areas may lead to potential congestion and queuing. Further investigations may explore the impact of the games rules with a higher rotation rate on students'physical activity during recess intervals, aiming to increase MVPA time for students. Additionally, this result also highlights the utilization of negative space during unstructured recess, and it is important for supervisors to divert and guide the pedestrian flow in an excessive influx of active area toward negative spaces. To increase the utilization of negative space, future research should explore the reasons why it is not being fully utilized, in order to take measures to better leverage negative space to promote physical activity during recess.

Another result is different from the original hypothesis, the intervention only increased the duration of VPA during unstructured recess in boys. The results of the study contradict earlier findings [16]. The possible reason is the existing different cognition of unstructured activities between boys and girls. Most boys regard it as an opportunity to participate in competitive games, while girls regard it as an opportunity to socialize with friends [49]. In detail, boys may be more inclined to choose equipment or facilities supporting high-intensity sports, while girls prefer green spaces or areas with lower overall exercise intensity during unstructured activities. The intervention design of the study is based on the existing equipment in school, with activities like jumping and agility ladder favoring higher intensity, which are more likely to attract boys. This possibility is further supported by the SOPLAY observation results in the study, which showed the number of boys participating in MVPA in the Z2 and Z4 areas was higher than that of girls after the intervention, suggesting that the intervention may have had a greater appeal to boys than to girls. These results indicate that future research may simultaneously increase the variety of activity environments in activity spaces for the health interests of all students, and explore the impact of students'PA.

Future implication

The results of the study highlight the potential for future research on enhancing PA through structured and unstructured recess. Interventions for structured activities provide students with clear guidance and rules through teachers, making structured recess particularly effective in enhancing physical activity levels for all students [8]. In contrast, interventions for unstructured activities, although not guaranteeing the same level of overall participation, have unique advantages. By creating more accessible and attractive spaces, they promote students'spontaneous participation in recess activities, encouraging them to engage in physical activity in a more autonomous and social manner [8]. This suggests that, in cases where school resources are limited, future recess interventions should choose between the two based on the functional goals of the recess activity and the school's educational objectives. For instance, if the goal is to increase the volume of recess activity while fostering self-organized exercise habits, unstructured recess may be the preferred intervention target. On the other hand, if the goal is to enhance PA across all students, structured recess should be chosen as the intervention focus.

Strength annd limitatio

The study is the first to use spatial analysis to objectively design activity space during recess, and provide a quantified recess intervention program in urban elementary schools with small size spaces. Most importantly, the material used in the study is from the existing equipment of the school, which is a low-cost intervention. However, the following limitations of the study should be acknowledged, and further exploration is needed in future research.

The first limitation lies in the immediate intervention. Although immediate interventions offer preliminary insights into feasibility and effectiveness, guiding the direction of future research, the lack of long-term intervention prevents the judgment of sustained effects, as well as the evaluation of intervention effects under different conditions [50], such as seasons, weather, etc. Subsequent studies with a duration exceeding six months are essential for a comprehensive evaluation of sustainability and for understanding the influence of various factors on its effectiveness. The second limitation is to try only 15 of minutes structured and unstructured recess intervention. Given the variations in recess activity rules across different countries, future research should explore the impact of intervention programs on students'physical activity with different recess durations. The third limitation relates to the physical activity behavior assessment days. Although some studies have employed single-day behavior observations and acclerometer tests between baseline and interventions, to minimize the risk of behavior observation and test chance, continuous observation days should be implemented in future studies. The fourth limitation is the case of observation. Although the case school in the study is a small school, optimizing space utilization is an effective strategy to enhance recess physical activity (PA) regardless of the school size. Future studies should include more cases from schools of different sizes, and compare the differences in intervention effects between schools of different sizes.

Conclusion

The study demonstrates that changing the activity type to Tabata exercise effectively enhances PA during structured recess for all elementary school students; adding game markings based on spatial analysis technology increases the overall number of students engaged in PA. These findings offer a cost-effective intervention program for promoting both structured and unstructured recess in small -sized urban schools. Further refinement and empirical validation of this intervention program are warranted for future research.

Abbreviations

PA	Physical activity
LPA	Light physical activity
MVPA	Moderate to vigorous physical activity
VPA	Vigorous physical activity
SOPLAY	System for Observing Play and Leisure Activity in Youth
IV	Integration value
PFS	pedestrian flow simulation

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12889-025-24451-z.

Supplementary material

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Authors' contributions

YY is responsible for data collection, analysis, and drafting the full article; DZ reviews the entire document, making revisions to the text and figures; YL proposes the research direction and conducts a review of the article. All authors read and approved the final manuscript.

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Data availability

Due to this study being a pilot study of intervention among primary school students in collaboration with a school in Suzhou Ping Zhi primary school, the data contains sensitive personal information that could identify the minors involved. Given the importance of protecting the privacy of participants, especially considering that the participants are underage students, we must ensure their personal information and privacy are strictly safeguarded. Therefore, in accordance with the agreement with the collaborating school and following relevant privacy protection laws and ethical standards, we are unable to publicly share the research data. We recognize that this may limit the transparency and verifiability of the data, but we believe that protecting the privacy and safety of the participants is paramount. All necessary measures have been taken to ensure the confidentiality of the data, and the presentation of the research findings does not disclose any personally identifiable information. For peers interested in further understanding the research methodology and findings, the detail data can be available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This intervention study is the part of Research on promoting sports literacy of children and adolescents, it has been approved by Institution of Review Board at Shanghai University of Sport, the full name of the ethics committee is Scientific Research Ethics Committee of Shanghai Sport University (Number: 102772022RT032), detail in additional material "Research Ethic supplementary file". Additional approvals were obtained from the Suzhou Pingzhi Experimental Primary School board and the president. All participating primary school students and their legal guardians have been fully informed about the nature, purpose, potential benefits, and possible risks of the study. Before starting any research activities, we have obtained written informed consent from all subjects and their legal guardians, ensuring their participation is based on full understanding and voluntary consent.

Consent for publication

Not applicable.

Competing Interests

The authors declare no competing interests.

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